

Creation Date 31-Jan-2011

Revision Date 09-Feb-2024

Revision Number 8

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: Tetrabutylammonium phosphate monobasic, 1M solution in ethanol  
Cat No. : 440060000; 440061000  
Molecular Formula C16 H36 N . H2 O4 P

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

#### Company

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

**Swiss distributor - Fisher Scientific AG**  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

E-mail address begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

# SAFETY DATA SHEET

Tetrabutylammonium phosphate monobasic, 1M solution in ethanol

Revision Date 09-Feb-2024

## CLP Classification - Regulation (EC) No 1272/2008

### Physical hazards

Flammable liquids

Category 2 (H225)

### Health hazards

Serious Eye Damage/Eye Irritation

Category 2 (H319)

### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Danger

### Hazard Statements

H225 - Highly flammable liquid and vapor

H319 - Causes serious eye irritation

### Precautionary Statements

P280 - Wear eye protection/ face protection

P264 - Wash face, hands and any exposed skin thoroughly after handling

P337 + P313 - If eye irritation persists: Get medical advice/attention

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

## 2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.2. Mixtures

| Component                                        | CAS No    | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008 |
|--------------------------------------------------|-----------|-------------------|----------|---------------------------------------------------|
| 1-Butanaminium, N,N,N-tributyl-, phosphate (1:1) | 5574-97-0 | EEC No. 226-947-1 | 38       | -                                                 |
| Ethyl alcohol                                    | 64-17-5   | 200-578-6         | 62       | Flam. Liq. 2 (H225)<br>Eye Irrit. 2 (H319)        |

| Component | Specific concentration limits | M-Factor | Component notes |
|-----------|-------------------------------|----------|-----------------|
|-----------|-------------------------------|----------|-----------------|

# SAFETY DATA SHEET

Tetrabutylammonium phosphate monobasic, 1M solution in ethanol

Revision Date 09-Feb-2024

|               | (SCL's)                |   |   |
|---------------|------------------------|---|---|
| Ethyl alcohol | Eye Irrit. 2 :: C>=50% | - | - |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | If symptoms persist, call a physician.                                                                                                           |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water.                                                                                     |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### 4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

### 4.3. Indication of any immediate medical attention and special treatment needed

|                           |                        |
|---------------------------|------------------------|
| <b>Notes to Physician</b> | Treat symptomatically. |
|---------------------------|------------------------|

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### Extinguishing media which must not be used for safety reasons

No information available.

### 5.2. Special hazards arising from the substance or mixture

Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Extremely flammable.

#### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Nitrogen oxides (NO<sub>x</sub>), Oxides of phosphorus.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

# SAFETY DATA SHEET

Tetrabutylammonium phosphate monobasic, 1M solution in ethanol

Revision Date 09-Feb-2024

## 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation.

## 6.2. Environmental precautions

Should not be released into the environment.

## 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Ensure adequate ventilation.

### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Flammables area. Keep away from heat, sparks and flame. Keep containers tightly closed in a dry, cool and well-ventilated place.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Class 3

**Switzerland - Storage of hazardous substances**

Storage class - SC 3  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### **Exposure limits**

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component     | European Union | The United Kingdom                                                                                               | France                                                                                                      | Belgium                                                       | Spain                                                                                              |
|---------------|----------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Ethyl alcohol |                | TWA: 1000 ppm TWA;<br>1920 mg/m <sup>3</sup> TWA<br>WEL - STEL: 3000 ppm<br>STEL: 5760 mg/m <sup>3</sup><br>STEL | TWA / VME: 1000 ppm<br>(8 heures).<br>TWA / VME: 1900<br>mg/m <sup>3</sup> (8 heures).<br>STEL / VLCT: 5000 | TWA: 1000 ppm 8 uren<br>TWA: 1907 mg/m <sup>3</sup> 8<br>uren | STEL / VLA-EC: 1000<br>ppm (15 minutos).<br>STEL / VLA-EC: 1910<br>mg/m <sup>3</sup> (15 minutos). |

# SAFETY DATA SHEET

Tetrabutylammonium phosphate monobasic, 1M solution in ethanol

Revision Date 09-Feb-2024

|  |  |  |                                                  |  |  |
|--|--|--|--------------------------------------------------|--|--|
|  |  |  | ppm.<br>STEL / VLCT: 9500<br>mg/m <sup>3</sup> . |  |  |
|--|--|--|--------------------------------------------------|--|--|

| Component     | Italy | Germany                                           | Portugal                     | The Netherlands                                                                         | Finland                                                                                                                                                        |
|---------------|-------|---------------------------------------------------|------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethyl alcohol |       | 200 ppm TWA MAK;<br>380 mg/m <sup>3</sup> TWA MAK | STEL: 1000 ppm 15<br>minutos | huid<br>STEL: 1900 mg/m <sup>3</sup> 15<br>minuten<br>TWA: 260 mg/m <sup>3</sup> 8 uren | TWA: 1000 ppm 8<br>tunteina<br>TWA: 1900 mg/m <sup>3</sup> 8<br>tunteina<br>STEL: 1300 ppm 15<br>minuutteina<br>STEL: 2500 mg/m <sup>3</sup> 15<br>minuutteina |

| Component     | Austria                                                                                                                                                              | Denmark                                                                                                                                         | Switzerland                                                                                                                                        | Poland                                     | Norway                                                                                                                                                                                |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethyl alcohol | MAK-KZGW: 2000 ppm<br>15 Minuten<br>MAK-KZGW: 3800<br>mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 1000 ppm 8<br>Stunden<br>MAK-TMW: 1900 mg/m <sup>3</sup><br>8 Stunden | TWA: 1000 ppm 8 timer<br>TWA: 1900 mg/m <sup>3</sup> 8<br>timer<br>STEL: 2000 ppm 15<br>minutter<br>STEL: 3800 mg/m <sup>3</sup> 15<br>minutter | STEL: 1000 ppm 15<br>Minuten<br>STEL: 1920 mg/m <sup>3</sup> 15<br>Minuten<br>TWA: 500 ppm 8<br>Stunden<br>TWA: 960 mg/m <sup>3</sup> 8<br>Stunden | TWA: 1900 mg/m <sup>3</sup> 8<br>godzinach | TWA: 500 ppm 8 timer<br>TWA: 950 mg/m <sup>3</sup> 8 timer<br>STEL: 625 ppm 15<br>minutter. value<br>calculated<br>STEL: 1187.5 mg/m <sup>3</sup> 15<br>minutter. value<br>calculated |

| Component     | Bulgaria                    | Croatia                                                                        | Ireland               | Cyprus | Czech Republic                                                                |
|---------------|-----------------------------|--------------------------------------------------------------------------------|-----------------------|--------|-------------------------------------------------------------------------------|
| Ethyl alcohol | TWA: 1000 mg/m <sup>3</sup> | TWA-GVI: 1000 ppm 8<br>satima.<br>TWA-GVI: 1900 mg/m <sup>3</sup><br>8 satima. | STEL: 1000 ppm 15 min |        | TWA: 1000 mg/m <sup>3</sup> 8<br>hodinách.<br>Ceiling: 3000 mg/m <sup>3</sup> |

| Component     | Estonia                                                                                                                                                       | Gibraltar | Greece                                       | Hungary                                                                                         | Iceland                                                                                                                                      |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Ethyl alcohol | TWA: 500 ppm 8<br>tundides.<br>TWA: 1000 mg/m <sup>3</sup> 8<br>tundides.<br>STEL: 1000 ppm 15<br>minutites.<br>STEL: 1900 mg/m <sup>3</sup> 15<br>minutites. |           | TWA: 1000 ppm<br>TWA: 1900 mg/m <sup>3</sup> | STEL: 3800 mg/m <sup>3</sup> 15<br>percekben. CK<br>TWA: 1900 mg/m <sup>3</sup> 8<br>órában. AK | TWA: 1000 ppm 8<br>klukkustundum.<br>TWA: 1900 mg/m <sup>3</sup> 8<br>klukkustundum.<br>Ceiling: 2000 ppm<br>Ceiling: 3800 mg/m <sup>3</sup> |

| Component     | Latvia                      | Lithuania                                                                                                  | Luxembourg | Malta | Romania                                                                                                                              |
|---------------|-----------------------------|------------------------------------------------------------------------------------------------------------|------------|-------|--------------------------------------------------------------------------------------------------------------------------------------|
| Ethyl alcohol | TWA: 1000 mg/m <sup>3</sup> | TWA: 500 ppm IPRD<br>TWA: 1000 mg/m <sup>3</sup><br>IPRD<br>STEL: 1000 ppm<br>STEL: 1900 mg/m <sup>3</sup> |            |       | TWA: 1000 ppm 8 ore<br>TWA: 1900 mg/m <sup>3</sup> 8 ore<br>STEL: 5000 ppm 15<br>minute<br>STEL: 9500 mg/m <sup>3</sup> 15<br>minute |

| Component     | Russia                                                          | Slovak Republic                                                               | Slovenia                                                                                                                               | Sweden                                                                                                                                                                            | Turkey |
|---------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Ethyl alcohol | TWA: 1000 mg/m <sup>3</sup> 2391<br>MAC: 2000 mg/m <sup>3</sup> | Ceiling: 1920 mg/m <sup>3</sup><br>TWA: 500 ppm<br>TWA: 960 mg/m <sup>3</sup> | TWA: 960 mg/m <sup>3</sup> 8 urah<br>TWA: 500 ppm 8 urah<br>STEL: 1000 ppm 15<br>minutah<br>STEL: 1920 mg/m <sup>3</sup> 15<br>minutah | Indicative STEL: 1000<br>ppm 15 minuter<br>Indicative STEL: 1900<br>mg/m <sup>3</sup> 15 minuter<br>TLV: 500 ppm 8 timmar.<br>NGV<br>TLV: 1000 mg/m <sup>3</sup> 8<br>timmar. NGV |        |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

# SAFETY DATA SHEET

Tetrabutylammonium phosphate monobasic, 1M solution in ethanol

Revision Date 09-Feb-2024

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                       | Acute effects local (Oral) | Acute effects systemic (Oral) | Chronic effects local (Oral) | Chronic effects systemic (Oral) |
|---------------------------------|----------------------------|-------------------------------|------------------------------|---------------------------------|
| Ethyl alcohol<br>64-17-5 ( 62 ) |                            | DNEL = 87 mg/kg bw/d          |                              |                                 |

| Component                       | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|---------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Ethyl alcohol<br>64-17-5 ( 62 ) |                              |                                 |                                | DNEL = 343mg/kg bw/day            |

| Component                       | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|---------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Ethyl alcohol<br>64-17-5 ( 62 ) | DNEL = 1900mg/m <sup>3</sup>     |                                     |                                    | DNEL = 950mg/m <sup>3</sup>           |

## Predicted No Effect Concentration (PNEC)

See values below.

## 8.2. Exposure controls

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material                                      | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|-----------------------------------------------------|-----------------------------------|-----------------|-------------|-----------------------|
| Natural rubber<br>Nitrile rubber<br>Neoprene<br>PVC | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

# SAFETY DATA SHEET

Tetrabutylammonium phosphate monobasic, 1M solution in ethanol

Revision Date 09-Feb-2024

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

## Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141

When RPE is used a face piece Fit Test should be conducted

## Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Physical State                          | Liquid                   |                                   |
| Appearance                              |                          |                                   |
| Odor                                    | No information available |                                   |
| Odor Threshold                          | No data available        |                                   |
| Melting Point/Range                     | No data available        |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | No information available |                                   |
| Flammability (liquid)                   | Highly flammable         | On basis of test data             |
| Flammability (solid,gas)                | Not applicable           | Liquid                            |
| Explosion Limits                        | No data available        |                                   |
| Flash Point                             | 14 °C / 57.2 °F          | Method - No information available |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| pH                                      | No information available |                                   |
| Viscosity                               | No data available        |                                   |
| Water Solubility                        | No information available |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Component                               | log Pow                  |                                   |
| Ethyl alcohol                           | -0.32                    |                                   |
| Vapor Pressure                          | No data available        |                                   |
| Density / Specific Gravity              | 0.895                    |                                   |
| Bulk Density                            | Not applicable           | Liquid                            |
| Vapor Density                           | No data available        | (Air = 1.0)                       |
| Particle characteristics                | Not applicable (liquid)  |                                   |

### 9.2. Other information

|                      |                                             |
|----------------------|---------------------------------------------|
| Molecular Formula    | C16 H36 N . H2 O4 P                         |
| Molecular Weight     | 339.45                                      |
| Explosive Properties | Vapors may form explosive mixtures with air |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

# SAFETY DATA SHEET

Tetrabutylammonium phosphate monobasic, 1M solution in ethanol

Revision Date 09-Feb-2024

## Hazardous Polymerization Hazardous Reactions

Hazardous polymerization does not occur.  
None under normal processing.

## 10.4. Conditions to avoid

Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition.

## 10.5. Incompatible materials

Strong oxidizing agents. Peroxides. Acids. Acid anhydrides. Acid chlorides. Ammonia.

## 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>). Oxides of phosphorus.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

##### (a) acute toxicity;

|            |                   |
|------------|-------------------|
| Oral       | No data available |
| Dermal     | No data available |
| Inhalation | No data available |

#### Toxicology data for the components

| Component     | LD50 Oral                                                    | LD50 Dermal | LC50 Inhalation                                                   |
|---------------|--------------------------------------------------------------|-------------|-------------------------------------------------------------------|
| Ethyl alcohol | LD50 = 10470 mg/kg<br>OECD 401 (Rat)<br>3450 mg/kg ( Mouse ) | -           | LC50 = 117-125 mg/l (4h)<br>OECD 403 (rat)<br>20000 ppm/10H (rat) |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

##### (d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

| Component                       | Test method                                       | Test species | Study result    |
|---------------------------------|---------------------------------------------------|--------------|-----------------|
| Ethyl alcohol<br>64-17-5 ( 62 ) | Mouse Ear Swelling Test (MEST)                    | mouse        | non-sensitising |
|                                 | OECD Test Guideline 429<br>Local Lymph Node Assay | mouse        | non-sensitising |

(e) germ cell mutagenicity; No data available

| Component                       | Test method                                   | Test species          | Study result |
|---------------------------------|-----------------------------------------------|-----------------------|--------------|
| Ethyl alcohol<br>64-17-5 ( 62 ) | AMES test<br>OECD Test Guideline 471          | in vitro<br>Bacteria  | negative     |
|                                 | Gene cell mutation<br>OECD Test Guideline 476 | in vitro<br>Mammalian | negative     |

(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen



# SAFETY DATA SHEET

Tetrabutylammonium phosphate monobasic, 1M solution in ethanol

Revision Date 09-Feb-2024

| (g) reproductive toxicity;      |                         | No data available            |                       |
|---------------------------------|-------------------------|------------------------------|-----------------------|
| Component                       | Test method             | Test species / Duration      | Study result          |
| Ethyl alcohol<br>64-17-5 ( 62 ) | OECD Test Guideline 416 | Oral / mouse<br>2 Generation | NOAEL = 13.8 g/kg/day |
|                                 | OECD Test Guideline 414 | Inhalation / Rat             | NOAEC =<br>16000 ppm  |

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Other Adverse Effects The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

## 11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

Ecotoxicity effects Contains a substance which is: Toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component     | Freshwater Fish                                            | Water Flea                                    | Freshwater Algae                           |
|---------------|------------------------------------------------------------|-----------------------------------------------|--------------------------------------------|
| Ethyl alcohol | Fathead minnow (Pimephales promelas) LC50 = 14200 mg/l/96h | EC50 = 9268 mg/L/48h<br>EC50 = 10800 mg/L/24h | EC50 (72h) = 275 mg/l (Chlorella vulgaris) |

| Component     | Microtox                                                                                                    | M-Factor |
|---------------|-------------------------------------------------------------------------------------------------------------|----------|
| Ethyl alcohol | Photobacterium phosphoreum: EC50 = 34634 mg/L/30 min<br>Photobacterium phosphoreum: EC50 = 35470 mg/L/5 min |          |

12.2. Persistence and degradability No information available

| Component                       | Degradability   |
|---------------------------------|-----------------|
| Ethyl alcohol<br>64-17-5 ( 62 ) | OECD 301E = 94% |

Degradation in sewage treatment plant Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential No information available

| Component     | log Pow | Bioconcentration factor (BCF) |
|---------------|---------|-------------------------------|
| Ethyl alcohol | -0.32   | No data available             |

12.4. Mobility in soil No information available .

# SAFETY DATA SHEET

Tetrabutylammonium phosphate monobasic, 1M solution in ethanol

Revision Date 09-Feb-2024

**12.5. Results of PBT and vPvB assessment** No data available for assessment.

**12.6. Endocrine disrupting properties**

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**12.7. Other adverse effects**  
**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

**13.1. Waste treatment methods**

**Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**European Waste Catalogue (EWC)**

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations.

**Switzerland - Waste Ordinance**

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO**

**14.1. UN number**

UN1170

**14.2. UN proper shipping name**

Ethanol solution

**14.3. Transport hazard class(es)**

3

**14.4. Packing group**

II

**ADR**

**14.1. UN number**

UN1170

**14.2. UN proper shipping name**

Ethanol solution

**14.3. Transport hazard class(es)**

3

**14.4. Packing group**

II

**IATA**

**14.1. UN number**

UN1170

**14.2. UN proper shipping name**

Ethanol solution

**14.3. Transport hazard class(es)**

3

**14.4. Packing group**

II

**14.5. Environmental hazards**

No hazards identified

# SAFETY DATA SHEET

Tetrabutylammonium phosphate monobasic, 1M solution in ethanol

Revision Date 09-Feb-2024

**14.6. Special precautions for user** No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                                        | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|--------------------------------------------------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| 1-Butanaminium, N,N,N-tributyl-, phosphate (1:1) | 5574-97-0 | 226-947-1 | -      | -   | -     | X    | -        | -    | -    |
| Ethyl alcohol                                    | 64-17-5   | 200-578-6 | -      | -   | X     | X    | KE-13217 | X    | X    |

| Component                                        | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|--------------------------------------------------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| 1-Butanaminium, N,N,N-tributyl-, phosphate (1:1) | 5574-97-0 | X    | ACTIVE                                        | X   | -    | X    | X     | -     |
| Ethyl alcohol                                    | 64-17-5   | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### Authorisation/Restrictions according to EU REACH

Not applicable

| Component                                        | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------------------------------------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1-Butanaminium, N,N,N-tributyl-, phosphate (1:1) | 5574-97-0 | -                                                                   | -                                                                             | -                                                                                                     |
| Ethyl alcohol                                    | 64-17-5   | -                                                                   | -                                                                             | -                                                                                                     |

#### Seveso III Directive (2012/18/EC)

| Component                                        | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|--------------------------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 1-Butanaminium, N,N,N-tributyl-, phosphate (1:1) | 5574-97-0 | Not applicable                                                                            | Not applicable                                                                           |
| Ethyl alcohol                                    | 64-17-5   | Not applicable                                                                            | Not applicable                                                                           |

#### Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

#### Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

#### National Regulations

# SAFETY DATA SHEET

Tetrabutylammonium phosphate monobasic, 1M solution in ethanol

Revision Date 09-Feb-2024

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

**WGK Classification** Water endangering class = 1 (self classification)

| Component     | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|---------------|---------------------------------------|-------------------------|
| Ethyl alcohol | WGK1                                  |                         |

| Component     | France - INRS (Tables of occupational diseases)      |
|---------------|------------------------------------------------------|
| Ethyl alcohol | Tableaux des maladies professionnelles (TMP) - RG 84 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                       | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|---------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Ethyl alcohol<br>64-17-5 ( 62 ) |                                                                                                                | Group I                                                                         |                                                                                             |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H319 - Causes serious eye irritation

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

**Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

**Physical hazards**

On basis of test data

**Health Hazards**

Calculation method

# SAFETY DATA SHEET

Tetrabutylammonium phosphate monobasic, 1M solution in ethanol

Revision Date 09-Feb-2024

Environmental hazards

Calculation method

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Creation Date 31-Jan-2011

Revision Date 09-Feb-2024

Revision Summary Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**